AURORA SDV Embedded Control Overview

Proteus + STM32 proof-of-concept architecture

Purpose

This document describes the embedded control portion of AURORA SDV implemented in Proteus using three STM32F401RE microcontrollers. The prototype demonstrates the control chain from battery protection, to vehicle-level decision making, to motor actuation and fault feedback.

Embedded Architecture

BMS → VCU → Motor Controller → PWM / Motor Stage

Motor Controller → health / fault feedback → VCU

Subsystem	Role in the current Proteus prototype
BMS	Reads battery-temperature input, determines NORMAL/WARNING/DERATE/SHUTDOWN state, drives green/red status LEDs, and sends the state to the VCU.
VCU	Reads accelerator, brake and SOC inputs plus ignition and charge switches; receives BMS status; determines OFF/READY/DRIVE/REGEN/CHARGE/FAULT; calculates final motor command.
Motor Controller	Receives DRIVE/REGEN/STOP command from the VCU, generates hardware PWM on TIM3_CH1, monitors motor-fault input and reports motor health back to the VCU.

Important Scope Clarifications

- The BMS in the current Proteus schematic does not calculate SOC and does not include a dedicated current sensor.
- SOC is represented by an input on the VCU.
- The ignition/start switch and charge switch are on the VCU.
- The BMS currently demonstrates battery-temperature protection plus status LEDs.
- Regenerative braking is represented in the VCU logic, but the simplified single-MOSFET motor stage does not physically return energy to the source.
- The detailed PMSM/FOC/EKF motor-control work belongs to the MATLAB/Simulink side of the project; the Proteus motor stage is the embedded integration proof-of-concept.

BMS Demonstration States

Temperature	BMS State	Torque Limit
< 30 °C	NORMAL	100%
30 °C to < 40 °C	WARNING	80%
40 °C to < 80 °C	DERATE	50%
≥ 80 °C	SHUTDOWN	0%

These thresholds are demonstration values chosen to make all states easy to reach during Proteus testing; they are not claimed as validated production battery limits.

Verified Normal-Drive Example

With BMS = NORMAL, accelerator = 50% and no active fault, the VCU produces a 50% motor command. The Motor Controller receives mode D with value 50 and applies approximately 50% PWM.

Software Used

Tool	Use
Proteus	Electrical and embedded co-simulation.
STM32CubeMX	Peripheral configuration for ADC, UART, GPIO and TIM3.
STM32CubeIDE	Firmware development, build and ELF generation.
MATLAB/Simulink	Higher-fidelity motor/control modelling and algorithm development.
Python / Dashboard	Planned live backend integration and HMI state display.